

11(a)(i)	electrons decelerate (on hitting target) so X-ray photons produced	B1
	range of decelerations	B1
	photon energy depends on (magnitude of) deceleration	B1
11(a)(ii)	$eV = \frac{hc}{\lambda}$	C1
	$\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{1.6 \times 10^{-19} \times 15000}$	C1
	$= 8.3 \times 10^{-11} m$	A1
	or $E = hf$ <u>and</u> $c = f\lambda$ <u>and</u> electron energy = eV or $E = hc / \lambda$ <u>and</u> electron energy = eV electron energy = $1.6 \times 10^{-19} \times 15000$ $= 2.4 \times 10^{-15}$	(C1)
	$\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2.4 \times 10^{-15}}$	(C1)
	$\lambda = 8.3 \times 10^{-11} m$	(A1)
11(b)(i)	$\mu = -\text{gradient or } \ln(I/I_0) = -\mu x$	C1
	(e.g. $2.08 / 10.0$) = 0.21 cm ⁻¹	A1

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Question	Answer	Marks
11(b)(ii)	$\ln(0.05) = -\mu x$	C1
	$x = \frac{\ln 0.05}{-\mu}$ <i>e.g. $x = 14 \text{ cm}$</i>	A1

Question	Answer	Marks
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